Rajnish Maurya|Roll No: DA24M015 | INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
M.Tech. (Data Science & Artificial Intelligence)	Indian Institute of Technology, Madras	8.47	2026
M.Tech (Computer Science & Engineering)	Indian Institute of Technology,Bhubaneswar	8.16	2023-2024
B.Tech (Computer Science and Engineering)	Indian Institute of Information Technology, Kalyani	8.18	2022
Class XII (UP Board)	P D B P I C A H B , Gorakhpur, UP	72.8	2017
Class X (CBSE)	Surmount International School, Gorakhpur, UP	8	2015

- Selected for Amazon Machine Learning Summer School 2025
- Received IndiaAI Fellowship 2025 by Central Government of India
- Secured AIR 300 in GATE 2024(Data Science and Artificial Intelligence) among 40k candidates.
- Secured AIR 1594 in GATE 2023(Computer Science and Engineering) among 1 lakh candidates
- Solved 400+ coding problems on Leetcode and GeeksforGeeks

PROFESSIONAL EXPERIENCE

Applied Research Intern, Bosch Global Software Technologies, Bangalore, India (May'25 – Jul'25)

Trajectory training using vision-language foundation models

- Developed trajectory prediction models from video and image frames using fine-tuned vision-language and multimodal models.
- Engaged in designing prototypes, conducting literature reviews, and evaluating model performance.
- Achieved an RMSE of approximately 3 in trajectory prediction tasks for 60 points(x,y,z)

PROJECTS

M.Tech Projects, Nirav Bhatt , Assistant Professor, Department of Data Science & AI, (Aug'25 - Present)

Investigating LLMs for Calculus Question generations at different Bloom's Taxonomy levels

- Evaluate the ability of existing maths-focused LLMs to generate and paraphrase Calculus-specific questions in zero-shot and few-shot settings.
- Benchmark RAG-based pipelines, with and without knowledge graph integration, for these tasks.
- Compare the performance of standalone LLMs and RAG-based approaches to identify strengths, limitations, and optimal configurations.

MLOps Course Project (Feb'25- May'25)

Twitter Sentiment Analysis – End-to-End MLOps Pipeline

- Designed and deployed an end-to-end ML pipeline for tweet sentiment classification using SVM with TF-IDF features.
- Containerized and orchestrated the inference API (FastAPI), monitoring stack (Prometheus, Grafana), and MLflow tracking server using Docker Compose.
- Implemented reproducible workflows with DVC and integrated data drift detection via Prometheus PushGateway, visualized in Grafana dashboards.

B.tech Projects, Dr. Uma Das, Associate Professor, Department of Computer Science, IIIT Kalyani (Jul'21-May'22)

Prediction of Mean Stratospheric Temperature Over Equator Using LSTM Based RNN

- Implemented an LSTM-based RNN to predict stratospheric temperatures at various pressure altitudes.
- Trained and tested the model using the ERA5 reanalysis dataset containing 51 years (1979–2019) of monthly mean zonal temperatures.
- Achieved temperature predictions for 2018–2019 using a model trained on data from 1979–2017.

B.tech Projects,Dr. Uma Das, Associate Professor, Department of Computer Science,IIIT Kalyani (Jul'21-May'22)

Forecast the anticipated rainfall precipitation at different specific latitudes and longitudes

- Developed a climate change forecasting model to predict precipitation rates at 0.25° resolution across the country using historical data from 1998–2019.
- Applied ARIMA, SARIMAX, and LSTM models to estimate rainfall for the following year at specific latitude–longitude coordinates.

CERTIFICATIONS

Jan'25-May'25	Introduction to Large Language Models, NPTEL
Dec'20-Jan'21	Machine Learning with Python, Verzeo
May'20-Jun'20	Data Science, Internshala

COURSEWORK

- Data Structures for Data Science
- Design and Analysis of Algorithms
- Operating System
- Database Management Systems
- Calculus, Linear Algebra
- Probability and Statistics
- Optimization
- Data Analytics Laboratory
- Machine Learning Operations Lab
- Foundation of Machine Learning
- Natural Language Processing
- Deep Learning
- Modern Computer Vision

SKILLS AND TOOLS

- **Languages:** Python, C++, SQL
- **Tools & Libraries:** Pytorch, Tensorflow, Scikit-learn, NLTK, Pandas, NumPy, Flask, Git,MLflow, DVC, Grafana, Docker

POSITION OF RESPONSIBILITIES

- **Teaching Assistant**, Foundation of Machine Learning (Jan–May 2025), IIT Madras